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Digital technology adoption and perceived barriers in music composition education among preservice music teachers in Ghanaian Colleges of Education

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Abstract

Digital technology has transformed music composition pedagogy in well-resourced educational contexts worldwide. In sub-Saharan Africa, however, the adoption of digital composition tools in preservice music teacher education is poorly understood. Limited indexed empirical literature appears to address preservice music teachers' perceptions of digital technology in music composition education within Ghanaian Colleges of Education directly. This qualitative study addressed three research questions: (1) How do preservice music teachers at a Ghanaian College of Education perceive digital technology in music composition education? (2) What barriers do they encounter in adopting digital composition tools? (3) To what extent do they and their institution feel prepared for technology integration in composition teaching? Three research questions were retained because each addresses a distinct dimension of adoption—perceptual, structural, and institutional—that a single question cannot capture adequately. We used an interpretivist qualitative design with 22 final-year preservice music teachers at Abetifi Presbyterian College of Education in Ghana's Eastern Region. Data were collected through semi-structured interviews, classroom observations, and curriculum document review, and were analysed using reflexive thematic analysis [6, 7] with NVivo 14. Five themes emerged. Participants recognised the pedagogical value of digital composition tools — particularly real-time aural feedback and creative revision — but all 22 reported severe infrastructural barriers: unreliable electricity, scarce devices, and limited internet access. Sixteen participants identified a cultural-epistemological misalignment between Western-designed software and Ghanaian musical traditions built on oral transmission, polyrhythmic structures, and communal creativity. Seventeen described weak curriculum integration of digital tools, and fourteen expressed low confidence in their own and their institution's preparedness for technology adoption. The findings extend the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al. [25]) by demonstrating that its facilitating conditions construct cannot function where baseline resource infrastructure is absent. We propose the Ghanaian Digital Music Education Adoption Framework (G-DMEAF), which adds resource facilitation



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and cultural-musical fit as prerequisite moderators to the standard UTAUT constructs. The study has implications for curriculum reform, teacher professional development, and digital education policy in Ghana and comparable sub-Saharan African settings.

Keywords Digital technology, Music composition education, Preservice music teachers, Ghana, Technology adoption, UTAUT, G-DMEAF, Cultural responsiveness, Colleges of Education

1 Introduction

Digital technology has changed music composition pedagogy in educational systems across Europe, North America, and East Asia. Software platforms such as MuseScore, Sibelius, and Finale enable students to hear their compositions in real time, revise without re-notation, and produce professional-quality scores without needing live performers [15, 17]. Digital audio workstations—Ableton Live, Logic Pro, and GarageBand—extend this further by enabling multi-track recording and sound manipulation at low cost [22]. As Calderón-Garrido et al. [8] demonstrated in Spain, preservice teachers trained with digital composition tools show stronger score presentation skills and greater pedagogical confidence than those trained in analogue methods alone.

Preservice music teachers need digital competence not only as composers but as future educators. Dorfman [13] and Clausen [9] both found that teacher education programmes which treat digital tools as peripheral rather than central produce graduates who lack the skills and pedagogical models to teach music with technology. In Ghana, the situation is more acute. Aikins and Akuffo [2] found that music instruction in Ghanaian Colleges of Education remained predominantly offline during the COVID-19 pandemic, constrained by electricity outages, limited device access, and insufficient internet bandwidth. Ofori-Atta [21] specifically identified weak digital preparation in Ghanaian music teacher education programmes, while Flolu [14] traced these structural deficits to a colonial educational legacy that privileged Western notational systems over indigenous oral and communal approaches.

The problem is not only infrastructural. Ghanaian musical traditions—characterised by polyrhythmic structures, oral transmission, call-and-response patterns, and communal creativity—are shaped by epistemological frameworks that differ fundamentally from those embedded in commercial notation software [19]. Agawu [1] argued that African musical imagination operates through rhythmic organisation and communal participation that cannot be reduced to the equal-tempered, individually sequenced workflows of Western notation tools. Nzewi [20] made a comparable case: indigenous African musical knowledge systems build meaning through communal practice and oral transmission, not through score-based individual authorship. For preservice teachers asked to use software that cannot represent the music they are trained to teach, technology adoption involves a dual burden: learning unfamiliar tools and translating between incompatible musical worlds.

Despite the importance of this topic, the literature on it is thin. Studies on ICT adoption in Ghanaian higher education [2, 3] address infrastructure barriers but do not focus on music or on cultural-epistemological fit. The broader music technology literature [8, 18, 24] is largely grounded in resource-adequate Western contexts. Applications of the Unified Theory of Acceptance and Use of Technology (UTAUT; [25]) to music teacher education [5, 18] have not examined how cultural misalignment between software

design and local musical epistemology moderates adoption. These gaps together identify the need for a contextually grounded empirical study of digital technology adoption in Ghanaian music teacher education.

This study addressed that gap through three research questions. Three were retained rather than two because each addresses a distinct and non-overlapping dimension of technology adoption: perception, barrier, and institutional readiness. These three dimensions correspond directly to the three layers of constraint—functional, structural, and cultural-institutional—that the literature and the data each identified independently. First: how do preservice music teachers at a Ghanaian College of Education perceive digital technology in music composition education? Second: what barriers do they encounter in adopting digital composition tools? Third: to what extent do preservice music teachers see themselves and their institution as prepared for digital technology integration in composition teaching?

The study is grounded in UTAUT [25] and proposes a contextually appropriate extension—the Ghanaian Digital Music Education Adoption Framework (G-DMEAF)—built from the data and from scholarship on African musical knowledge systems. The remainder of this paper is structured as follows: Sect. 2 reviews the relevant literature; Sect. 3 presents the theoretical framework and G-DMEAF; Sect. 4 describes the methodology; Sect. 5 reports findings by theme, with explicit mapping to UTAUT constructs; Sect. 6 discusses findings in relation to prior literature; and Sect. 7 states the conclusions.

2 Literature review

2.1 Digital technology in music composition pedagogy

Research from resource-adequate settings establishes consistent findings about the pedagogical value of digital composition tools. Notation software provides immediate aural feedback, supports rapid revision, and produces professional-quality output [15, 17]. Calderón-Garrido et al. [8] found that structured digital composition training improved score production skills and pedagogical confidence in Spanish preservice music teachers. Reese et al. [22] documented how preservice teachers using iPads in United States secondary general music classrooms reported greater student engagement in composition and performance. Tang [24] observed that evolving digital platforms are reshaping music teaching strategies in ways that require continuous pedagogical adaptation from educators.

These gains depend on adequate preparation and support. Clausen [9] found that many teacher preparation programmes treat digital tools as supplementary, producing graduates who cannot model or assess digitally created compositions. Dorfman [13] identified wide variation across United States music teacher education programmes in how digital tools are integrated, with many treating technology as optional rather than required. Haleem et al. [16] showed that digital tool effectiveness in education is contingent on prior teacher exposure and sustained institutional support. Reese et al. [22] also noted that over-reliance on software playback functions can reduce aural skill development—a caveat relevant to any curriculum that prioritises software access without pedagogical scaffolding.

2.2 Technology adoption in preservice music teacher education

The UTAUT model [25], developed across four studies with 215 participants, identifies four constructs predicting technology acceptance: performance expectancy (belief that the technology will improve job performance), effort expectancy (perceived ease of use), social influence (perception that important others endorse use), and facilitating conditions (availability of organisational and technical infrastructure). Venkatesh et al. [26] extended the model into UTAUT2, adding hedonic motivation, price value, and habit to address consumer-context adoption, improving variance explanation in behavioural intention from 56 to 74%.

UTAUT has been applied to music teacher education in two notable studies. Bauer and Dammers [5] applied UTAUT constructs in a national survey of United States music teachers and found facilitating conditions—access to hardware, software, and technical support—to be the strongest predictor of technology integration. Merrick and Joseph [18] applied UTAUT to Australian music educators during the COVID-19 pandemic and found that performance expectancy—beliefs about digital tools' pedagogical usefulness—was the key adoption predictor. Neither study examined contexts in which baseline infrastructure (electricity, devices, internet) cannot be assumed, and neither addressed cultural-epistemological alignment between software design and local musical traditions.

Recent work signals that UTAUT requires contextual extension when applied outside Western organisational settings. Ateş and Polat [4] found that institutional and cultural context substantially moderated humanoid robot adoption in Turkish science education, even in a comparatively resource-adequate setting. Ren and Wu [23] found that cultural attitudes towards artificial intelligence shaped adoption more strongly than functional beliefs in higher education settings. These findings support the need for a context-sensitive extension of UTAUT in Ghanaian music teacher education.

2.3 Barriers to technology adoption in Ghanaian higher education

Structural obstacles to technology adoption in Ghanaian higher education are well documented. Aikins and Akuffo [2] found that Colleges of Education during the COVID-19 pandemic faced electricity outages, poor internet bandwidth, inadequate device-to-student ratios, and prohibitive software licensing costs. Akon-Yamga et al. [3] reported that teachers and students across Ghanaian STEM schools consistently identified poor infrastructure as the primary constraint on technology-enhanced learning. Ofori-Atta [21] argued that Ghanaian music teacher preparation had not kept pace with digital competency expectations because institutional policies to support technology integration remained underdeveloped.

Beyond infrastructure, the cultural context of Ghanaian music education poses a distinct adoption barrier. Nketia [19] described Ghanaian musical traditions—including the polyrhythmic bell patterns of Ewe drumming and the communal call-and-response structures of Akan music—as knowledge systems built through oral transmission and social participation rather than visual notation. Agawu [1] demonstrated that African musical imagination is shaped by rhythmic complexity, timbral variety, and the inseparability of music from community practice. Nzewi [20] argued explicitly that indigenous African musical arts must be studied and taught through frameworks that honour their

own epistemological roots. These arguments suggest that digital tool adoption in Ghanaian music education is not only a question of access but of cultural relevance.

2.4 Gaps in the literature

Three gaps motivate this study. First, limited empirical attention has been given to pre-service music teachers' perceptions of digital composition tools in Ghanaian Colleges of Education specifically; based on the literature reviewed for this study, no indexed study appears to address this focus directly. Second, existing UTAUT applications in music education have not examined how cultural misalignment between technology design and local musical epistemology moderates adoption intention. Third, the UTAUT facilitating conditions construct assumes a baseline of infrastructure that cannot be assumed in Ghanaian public Colleges of Education. This study was designed to address all three gaps.

3 Theoretical framework

3.1 UTAUT and its music education applications

This study is grounded in UTAUT [25]. UTAUT proposes four core constructs that predict behavioural intention to use technology: performance expectancy (the belief that using the technology will improve job performance), effort expectancy (the perceived ease of use), social influence (the degree to which important others are seen to support use), and facilitating conditions (the perceived availability of the organisational and technical infrastructure to support use). Experience, voluntariness, gender, and age moderate these relationships. The model has been validated across diverse educational contexts and has been applied specifically to music teacher education by Bauer and Dammers [5] and Merrick and Joseph [18].

UTAUT was selected for this study because its constructs map directly onto the phenomena of interest: preservice music teachers' beliefs about digital tools' pedagogical value, their confidence in using them, the social norms around technology at their institution, and the material conditions that enable or constrain use. The UTAUT constructs therefore served as sensitising concepts informing both the interview guide design and the second cycle of deductive coding during data analysis.

3.2 Limitations of UTAUT in the Ghanaian context

Two limitations of UTAUT are particularly relevant here. First, the facilitating conditions construct was developed in organisational contexts where functional infrastructure is baseline. In Ghanaian Colleges of Education, electricity, devices, and internet access cannot be assumed. A construct that treats infrastructure as a moderating facilitator misrepresents a context where its absence is the primary barrier. UTAUT2 [26] added hedonic motivation, price value, and habit to address consumer-context adoption, but retained the same infrastructure assumptions and did not introduce cultural-epistemological constructs.

Second, neither UTAUT nor UTAUT2 accounts for alignment between the epistemological assumptions of technology design and the musical knowledge systems of the teachers using it. Western notation software encodes assumptions about individual authorship, equal temperament, and visual score organisation. These diverge from the oral transmission, polyrhythmic structures, and communal creativity that characterise

Ghanaian musical traditions [1, 19, 20]. This epistemological gap is not captured by any existing UTAUT construct.

3.3 The Ghanaian Digital Music Education Adoption Framework (G-DMEAF)

G-DMEAF was developed inductively from the data of this study and deductively from the scholarship reviewed above. It is important to note at the outset that G-DMEAF is a preliminary conceptual framework generated from a single-site qualitative case study with 22 participants. It is not a formally tested or validated model. Future quantitative, multi-site research will be necessary to assess the predictive power of its constructs and to determine how far findings from this context transfer to other Ghanaian Colleges of Education or comparable sub-Saharan African settings. The framework retains all four core UTAUT constructs and introduces two additional moderators specific to Ghanaian and comparable sub-Saharan African contexts.

Resource facilitation refers to the availability and reliability of the material conditions necessary for meaningful technology engagement: consistent electricity, functioning devices, internet connectivity, affordable software, and technical support. Unlike UTAUT's facilitating conditions construct—which treats infrastructure as one moderating variable alongside social and organisational support—resource facilitation in G-DMEAF is conceptualised as a prerequisite. Where resource facilitation is absent, performance expectancy and social influence have no practical pathway to generate behavioural intention or actual use. Price value from UTAUT2 is subsumed within this construct, as software cost is one dimension of resource availability.

Cultural-musical fit refers to the degree of alignment between the epistemological assumptions of digital composition tools and the musical traditions that preservice teachers are expected to teach. Cultural-musical fit captures the fact that technology adoption is not only a functional decision—it is a cultural and pedagogical one. A teacher who cannot use a tool to represent the music their community practises faces a barrier that is neither infrastructural nor attitudinal in the conventional UTAUT sense; it is epistemological. This construct is new to UTAUT-based frameworks and emerged directly from the data reported in Sect. 5.

Figure 1 presents the G-DMEAF conceptual diagram. Resource facilitation and cultural-musical fit appear on the left as prerequisite moderators—conditions that must be sufficiently present before the four core UTAUT constructs (centre column) can generate behavioural intention and use behaviour (right). Experience and voluntariness remain as contextual moderators below, consistent with the original UTAUT. G-DMEAF differs from UTAUT2 in two ways: it elevates infrastructure from moderator to prerequisite status, and it introduces cultural-musical fit as a new epistemological moderator with no equivalent in any existing UTAUT extension. The framework is represented as a linear flow diagram, reflecting a left-to-right causal sequence: prerequisite conditions must be sufficiently present before the core UTAUT constructs become operative, which in turn generate behavioural intention and observed use behaviour. This linear representation should be read as a conceptual model of necessary conditions and their downstream relationships, not as a structural equation or fully elaborated causal network. The single-directional arrows indicate precedence and gating, not reciprocal feedback loops, though feedback dynamics may warrant investigation in future quantitative research.

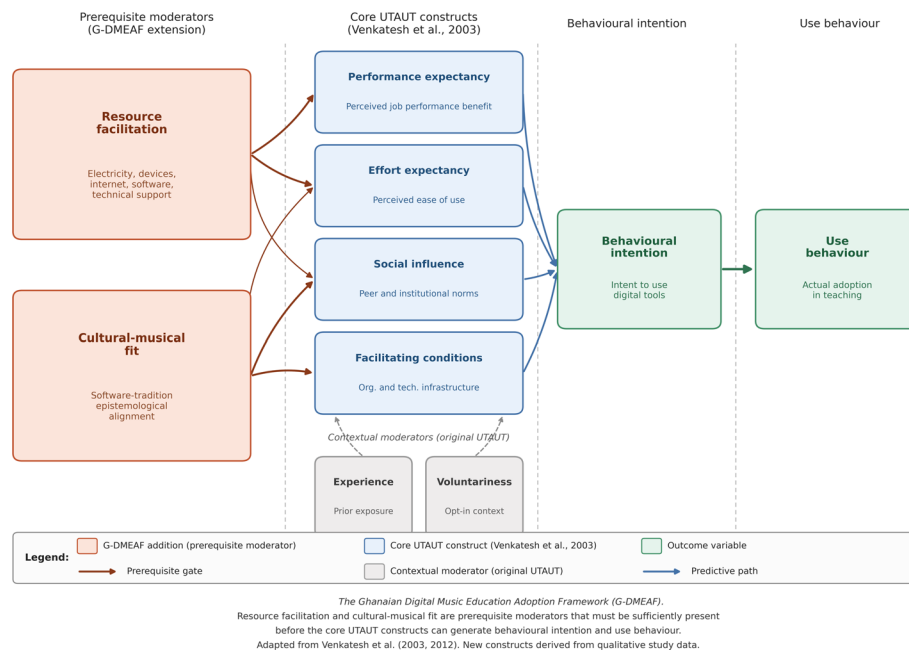


Fig. 1 The Ghanaian Digital Music Education Adoption Framework (G-DMEAF). Left column: resource facilitation and cultural-musical fit as prerequisite moderators. Centre column: four core UTAUT constructs (performance expectancy, effort expectancy, social influence, facilitating conditions). Right column: behavioural intention → use behaviour. Below: experience and voluntariness as contextual moderators. Bold arrows from resource facilitation and cultural-musical fit indicate prerequisite gating—these must be sufficiently present before UTAUT constructs become salient

4 Methodology

4.1 Research design and paradigm

We used a qualitative, interpretivist single-site case study design [11]. An interpretivist paradigm is appropriate here because technology adoption decisions in Ghanaian music teacher education are shaped by subjective perceptions, contextual meanings, and socially constructed understandings that cannot be captured through measurement alone. The qualitative design is justified because the research questions seek to understand how preservice music teachers make sense of digital composition tools in a specific institutional and cultural context that has not previously received empirical attention. This study was conducted at one College of Education; findings describe conditions at this site and are not generalised statistically to other Ghanaian institutions.

4.2 Researcher positionality

The first author is a music lecturer at Abetifi Presbyterian College of Education and was known to all participants. This proximity enabled deep rapport and candid accounts of institutional barriers that an external researcher might not have accessed. It also introduced the risk of social desirability bias. To manage this risk, we used four strategies throughout the study: anonymous data collection with pseudonyms assigned before data collection began; member checking with 12 participants after initial themes were developed; peer debriefing with two external researchers in music education and qualitative

methods; and a reflexive journal maintained by the first author throughout, documenting assumptions, interpretive decisions, and moments where positionality may have shaped interpretation. Journal entries were discussed in peer debriefing sessions and used to refine the analysis.

4.3 Research setting

The study was conducted at Abetifi Presbyterian College of Education, located in the Kwahu East District of Ghana’s Eastern Region. The college is one of 46 public Colleges of Education preparing teachers for basic schools across Ghana, offering a four-year Bachelor of Education degree with music as a specialisation. Composition instruction has historically relied on handwritten staff notation, tonic solfa systems, and live demonstration. Digital tools have not been formally integrated into the music curriculum. The college shares infrastructure constraints common to Ghanaian public Colleges of Education: intermittent electricity supply, limited internet bandwidth, and a single computer laboratory shared across all departments.

4.4 Participants and sampling

Participants were 22 final-year preservice music teachers selected through purposive sampling. Two criteria guided selection: completion of at least one semester of music composition coursework, and variation in prior digital technology experience to maximise diversity of perspectives. All participants were full-time Bachelor of Education students with music as their subject specialism. Part-time students and those without any composition coursework were excluded. Participation was voluntary, and no lecturer or department head was present during recruitment or data collection to minimise institutional pressure.

Table 1 presents the demographic characteristics of the 22 participants. Given the qualitative design of the study, a single consolidated demographic table is provided below.

4.5 Data collection

We collected data from January to March 2026 using three complementary methods. Semi-structured interviews were conducted with all 22 participants, each lasting 45 to 70 minutes. The interview guide comprised 18 questions organised around six domains:

Table 1 Demographic characteristics of the 22 preservice music teacher participants

	<i>n</i>	%
Gender		
Female	13	59.1
Male	9	40.9
Age range		
21–24 years	15	68.2
25–29 years	7	31.8
Prior digital technology experience		
None	6	27.3
Basic (occasional use)	11	50.0
Moderate (regular use)	5	22.7
Total	22	100.0

Percentages are descriptive demographics only and do not imply statistical generalisability. The total of 22 confirms consistency with the purposive sample. Data are presented for participant transparency, not for inferential analysis

Table 2 Descriptive frequency summary of participant demographic variables ($N=22$)

Variable	Category	f	%	Cumulative %
Gender	Female	13	59.1	59.1
	Male	9	40.9	100.0
Age range	21–24 years	15	68.2	68.2
	25–29 years	7	31.8	100.0
Prior digital experience	None	6	27.3	27.3
	Basic	11	50.0	77.3
	Moderate	5	22.7	100.0

Percentages may not sum to 100 due to rounding. This table presents descriptive statistics only; the qualitative nature of the study precludes inferential analysis

f: frequency; SD: standard deviation

(a) prior experience with digital composition tools (performance expectancy probe); (b) perceived ease of use and confidence with software (effort expectancy probe); (c) peer and lecturer encouragement to use digital tools (social influence probe); (d) available resources, electricity, devices, and internet access (facilitating conditions and resource facilitation probe); (e) cultural and musical identity in relation to digital tools—whether software can represent the music participants teach (cultural-musical fit probe); and (f) perceived institutional readiness for technology integration (preparedness probe, addressing Research Question 3). Sample questions included: ‘What do you think digital composition software could offer you as a future music teacher?’; ‘How confident do you feel using software like MuseScore?’; ‘Do your lecturers encourage you to use digital tools in composition?’; ‘What resources would you need to make meaningful use of digital composition software?’; and ‘Do you think software like MuseScore can represent the music you are preparing to teach in Ghanaian schools?’. Interviews were conducted in English; Twi phrases were translated by a bilingual research assistant and verified by the first author. All interviews were audio-recorded and transcribed verbatim Table 2.

Four classroom observations were conducted in composition teaching sessions, each lasting 60 to 90 min. Observation notes recorded instructional methods, technology use or absence, student engagement, and resource availability. We also reviewed the college’s music course syllabi, composition assessment rubrics, and institutional ICT policy documents to understand formal curriculum expectations regarding digital technology in composition teaching.

4.6 Data analysis

Data were analysed using reflexive thematic analysis [6, 7]. This approach was chosen because it allows researchers to identify patterns across a data set while remaining attentive to context, researcher subjectivity, and the socially constructed nature of meaning—properties consistent with the study’s interpretivist paradigm. The analysis was abductive: coding began inductively in the first cycle, drawing directly from the data, and was followed in the second cycle by a deductive mapping of codes against UTAUT constructs to identify alignment and divergence. This abductive approach allowed themes to emerge from the data while remaining theoretically anchored.

Analysis proceeded in six phases [7]. Phase 1: repeated reading of all transcripts, observation notes, and documents to achieve data familiarity. Phase 2: systematic generation of initial codes using NVivo 14 software, first inductively, then deductively mapped to UTAUT constructs. Consistent with reflexive thematic analysis, the first and third

authors independently familiarised themselves with a 20% random subsample of transcripts and discussed their initial codes; these discussions were used to deepen reflexive engagement with the data rather than to achieve consensus as a reliability measure, in keeping with Braun and Clarke's [7] position that researcher subjectivity is an analytic resource rather than a problem to eliminate. Phase 3: grouping related codes into candidate themes. Phase 4: reviewing and refining themes against the full data set. Phase 5: defining and naming each theme. Phase 6: writing the final analysis.

Trustworthiness was established through four strategies: (1) triangulation of interview, observation, and document data throughout analysis; (2) member checking, in which 12 participants selected by maximum variation across gender and technology experience reviewed summaries of emerging themes and commented on their resonance and accuracy; (3) peer debriefing at two points during analysis with two external researchers in music education and qualitative methodology; and (4) the first author's reflexive journal, used to examine positionality-related interpretive risks throughout the study.

4.7 Ethical considerations

Ethical approval was granted by the Institutional Review Board of Abetifi Presbyterian College of Education (APCE/IRB/2026/002). The study was conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent before data collection, with explicit assurance of their right to withdraw at any point without penalty. Pseudonyms (P1–P22) are used throughout; all identifying information has been removed from quotations and observation notes. The authors declare no competing interests. Participants consented to the publication of anonymised quotations.

5 Findings

Reflexive thematic analysis produced five themes that collectively address the three research questions. Themes 1 and 2 address Research Question 1 (perceptions) and Research Question 2 (barriers). Theme 3 addresses both Research Question 1 and Research Question 2. Theme 4 addresses Research Question 2. Theme 5 addresses Research Question 3 (institutional preparedness). Each theme is mapped to the corresponding UTAUT or G-DMEAF construct to clarify the theoretical significance of the data.

5.1 Theme 1: perceived pedagogical benefits of digital composition tools

UTAUT construct: Performance expectancy. This theme reflects participants' beliefs about the pedagogical value of digital tools—their performance expectancy—and addresses Research Question 1.

Fifteen of the 22 participants described immediate aural feedback as the most meaningful benefit of digital composition tools. Participants contrasted this with analogue composition, in which written work cannot be heard until a performer is available. P14 described the experience directly: 'When I write on paper I have to wait until I can play it on the keyboard or ask someone to sing it. With MuseScore, I press play and hear everything immediately. It helps me fix problems faster.' Sixteen participants described digital tools as enabling creative experimentation—the ability to try harmonic or rhythmic variations without rewriting an entire score. P7 said: 'I can try a different rhythm and if it does not work, I change it without starting again. That is something I cannot do

on paper.' Twelve participants mentioned improved score legibility and professional presentation as a benefit in assessment and professional development contexts.

Classroom observations provided partial contextual support for these accounts. In the one session where individual computers were briefly accessible for a portion of the session, students engaged more actively with composition tasks and asked more questions about harmonic alternatives than in sessions relying solely on handwritten notation.

5.2 Theme 2: infrastructural and resource barriers

G-DMEAF construct: Resource facilitation (prerequisite). This theme reflects the resource facilitation prerequisite introduced in G-DMEAF and directly addresses Research Question 2. It demonstrates why facilitating conditions in the original UTAUT cannot function as a moderator when infrastructure is absent at the foundational level.

All 22 participants identified infrastructural barriers as the primary obstacle to digital composition tool adoption. Electricity outages were reported by 20 participants as frequent enough to make sustained digital work impractical. P3 said: 'The light can go at any time. Sometimes it goes in the morning and does not come back until evening. Any work you have not saved is lost.' Nineteen participants described internet connectivity as too unreliable for cloud-based tools, software updates, or online tutorials. Eighteen participants reported having no personal laptop or tablet and described sharing a single outdated computer laboratory with all departments in the college. Fifteen participants identified licensed software costs (Sibelius, Logic Pro) as prohibitive and reported using free open-source alternatives such as MuseScore when access was available. Nine participants noted the absence of any technical support person capable of assisting with music software specifically.

Observation data confirmed these reports. In three of the four observed composition sessions, electricity supply was interrupted at least once, and two sessions were conducted entirely without electricity. In all four sessions, no computer was used for composition; instruction proceeded with handwritten notation, a piano, and verbal demonstration.

5.3 Theme 3: cultural and epistemological misalignment

G-DMEAF construct: Cultural-musical fit (prerequisite). This theme reflects the cultural-musical fit construct introduced in G-DMEAF and addresses both Research Questions 1 and 2. It is the most theoretically original finding of this study.

Sixteen participants articulated a gap between what digital composition software can represent and the music they are trained to teach. The remaining six participants did not directly address this misalignment; five of them had no prior digital experience and framed their concerns exclusively around infrastructure and access, while one participant expressed a broadly positive orientation towards Western notation software and did not identify cultural incongruence. This variation is noted, though it does not diminish the salience of the cultural-epistemological theme given its prominence across the majority of accounts. Participants who did articulate misalignment consistently described Western notation software as built around assumptions (equal temperament, individual composition, and linear score organisation) that do not map onto Ghanaian musical traditions. P11 said: 'The software wants you to put every sound into a box. But the music I know does not fit into those boxes. The timing, the layers, the way people

play together—the software cannot hold all of that.’ P19 described the pedagogical consequence directly: ‘When I use MuseScore to write Akan music, I have to change the music to fit the software. I am not writing our music anymore. I am writing something that looks like our music but sounds wrong.’

Eleven participants identified specific software limitations: the inability to notate hemirhythm and polyrhythmic layering accurately, the absence of tuning systems that include Ghanaian inflections, and the lack of group composition modes that reflect communal musical practice. P4 suggested: ‘If the software could let more than one person compose at the same time, the way we play together, that would change everything.’ Classroom observations confirmed these perceptions: in two of the four observed sessions, composition tasks used Western tonal examples with no reference to Ghanaian musical forms, oral transmission practices, or communal composition approaches.

5.4 Theme 4: curriculum and pedagogical readiness gaps

UTAUT construct: Facilitating conditions (curriculum dimension). This theme extends the facilitating conditions construct to include curriculum readiness as a distinct institutional barrier and addresses Research Question 2.

Seventeen participants described the music curriculum as providing minimal systematic guidance on digital composition software. P2 said: ‘We study music theory and traditional notation for three years. Digital technology is mentioned maybe once or twice. How am I supposed to teach it if I have never been taught how to use it properly?’ Thirteen participants described the curriculum’s emphasis on handwritten notation and tonic solfa as making digital tools peripheral to their training. Nineteen participants reported the absence of any formal rubric or assessment framework for digitally created compositions, leaving students and teachers uncertain about how to evaluate software-produced work.

Observation data corroborated these accounts. In all four observed composition sessions, instruction relied exclusively on handwritten notation, the piano, and verbal demonstration. No digital composition software was used in any session, even in cases where the syllabus listed composition as the topic. Curriculum document review confirmed that ICT integration was mentioned in the college’s general ICT policy but was not operationalised in any music course syllabus.

5.5 Theme 5: low institutional self-efficacy and preparedness

UTAUT construct: Social influence and effort expectancy (institutional dimension). This theme reflects low social influence—few institutional norms endorsing technology use — and low effort expectancy — participants’ doubt about their own capacity to learn and use digital tools effectively. It directly addresses Research Question 3.

Fourteen participants expressed doubt about whether they or their institution were ready to integrate digital composition tools into teaching. P16 said: ‘I do not feel prepared. I have used MuseScore twice. I cannot teach something I barely know myself.’ P20 described the institutional dimension: ‘The college has not trained us. There are no workshops, no demonstrations, no one to show us how. Even if the electricity came and we all had computers, we would not know what to do with them.’ P8 connected low preparedness to low motivation: ‘If I know I will not be supported, why try? I will just do what I know works.’ Seven participants noted that lecturers themselves appeared

unfamiliar with digital composition tools, which further reduced participants' confidence that the institution had any coherent plan for technology integration. Classroom observations corroborated this institutional dimension: across all four observed sessions, no digital composition tool was modelled or referenced by the lecturer, and composition pedagogy proceeded entirely through conventional means, providing no observable institutional signal that digital integration was expected or valued.

6 Discussion

6.1 Pattern summary

The five themes reveal that preservice music teachers at Abetifi Presbyterian College of Education recognise genuine pedagogical value in digital composition tools — particularly real-time aural feedback and creative revision—but encounter three distinct layers of constraint that prevent this recognition from translating into adoption. The first layer is physical: electricity, devices, and internet access are chronically scarce. The second layer is cultural-epistemological: available software cannot represent the musical traditions participants are prepared to teach. The third layer is institutional: the curriculum does not develop digital competencies, there is no assessment framework for digitally produced work, and neither students nor lecturers have a clear pathway into digital composition practice.

6.2 Interpretation against prior literature

The infrastructure findings confirm and extend Aikins and Akuffo [2], who documented electricity and device deficits in Ghanaian Colleges of Education during the COVID-19 pandemic. Where Aikins and Akuffo described these as crisis-specific disruptions, this study found them as chronic, structural conditions. The severity—all 22 participants identifying infrastructure as primary barrier—aligns with Akon-Yamga et al. [3] across Ghanaian STEM education and with Ofori-Atta [21] in Ghanaian music teacher preparation. Infrastructure deficit in Ghanaian Colleges of Education is systemic, not exceptional.

The curriculum findings extend Dorfman [13], who showed that many United States music teacher education programmes treat digital tools as optional. At Abetifi, digital tools are not optional—they are absent from the curriculum entirely. This aligns with Clausen [9], who found that preservice music teachers in under-resourced programmes lacked both the skills and pedagogical models to teach with technology. Where Merrick and Joseph [18] found that Australian educators' performance expectancy was the key adoption predictor, this study shows that at Abetifi, performance expectancy exists—participants clearly articulated the tools' value—but cannot become practice without institutional curriculum support.

The cultural misalignment findings are the most theoretically original contribution of this study. Prior UTAUT applications in music education [5, 18] and in other education contexts [4, 23] have not examined how cultural-epistemological alignment between software design and local musical knowledge systems moderates adoption. Agawu [1] showed that African musical imagination is shaped by rhythmic complexity and communal participation that resist Western notation's categories. Nketia [19] documented that Ghanaian traditions—from Ewe polyrhythm to Akan call-and-response—are built through oral transmission and social practice. Nzewi [20] argued explicitly that African

indigenous musical arts require pedagogical frameworks that honour their own epistemological roots. This study shows that preservice teachers experience this mismatch not as an abstract cultural concern but as a practical barrier to using software to plan and teach lessons. For instance, the hemirhythmic layering characteristic of Ewe bell patterns in the Agbekor tradition—where multiple interlocking rhythmic lines create a composite that is inseparable from communal performance practice—cannot be accurately notated in standard Western staff notation, let alone in notation software that enforces equal temperament and single-voice staff lines. Akan call-and-response forms, which derive their compositional logic from communal improvisation and social context rather than from individual authorship or fixed scores, present a comparable challenge: the compositional act itself is distributed and situated in ways that individual notation software was not designed to support. These examples illuminate precisely why cultural-musical fit, as proposed in G-DMEAF, is a necessary construct for understanding technology adoption in Ghanaian music teacher education. This analysis is also consistent with decolonial educational perspectives [27, 28], which argue that imported educational technologies and curricula carry epistemological assumptions that can marginalise indigenous knowledge systems. Digital music composition software, as it currently exists, largely reflects Western European tonal and notational epistemology. Meaningful technology integration in Ghanaian music teacher education therefore requires not only resource provision but a critically reflexive curriculum that positions indigenous Ghanaian musical knowledge as central rather than peripheral.

The institutional self-efficacy findings extend Haleem et al. [16], who demonstrated that digital literacy adoption depends heavily on institutional support structures. The absence of those structures at Abetifi—no professional development, no software training, no assessment frameworks for digital composition—creates a low-efficacy environment in which even positively disposed individuals have no pathway to build digital composition competence.

6.3 Theoretical implications: UTAUT and G-DMEAF

The findings have direct implications for how UTAUT should be applied in under-resourced educational contexts. The facilitating conditions construct in Venkatesh et al. [25] treats infrastructure as a moderating variable—one factor among several that shape adoption intention. The findings here show that in contexts of chronic resource scarcity, infrastructure is better conceptualised as a prerequisite rather than a moderator. Where electricity, devices, and internet are absent, performance expectancy and social influence are effectively inert: a teacher cannot act on positive beliefs about a tool they cannot access. Price value from UTAUT2 [26] is relevant here—fifteen participants cited software cost as a barrier—but UTAUT2 still treats this as a moderator rather than a precondition.

G-DMEAF addresses both limitations. By positioning resource facilitation as a prerequisite and cultural-musical fit as an epistemological moderator without precedent in existing UTAUT extensions, the framework captures the two dimensions of technology adoption in Ghanaian music teacher education that UTAUT and UTAUT2 cannot. G-DMEAF is proposed as a preliminary conceptual framework generated from qualitative findings at a single site. It is not a formally tested model. Future quantitative research

should validate its constructs and assess their predictive power across a larger, multi-site sample.

6.4 Practical implications

For curriculum policy, incremental approaches—occasional workshops or isolated software demonstrations—are insufficient. Music teacher education curricula in Ghanaian Colleges of Education need systematic digital competency development, from foundational software skills through pedagogical application in composition, performance, and assessment. The Ghana Tertiary Education Commission and the National Teaching Council should review whether music education standards reflect digital composition competencies.

For technology developers, the cultural-musical fit findings make a direct case for software that can represent Ghanaian and broader African musical traditions: support for polyrhythmic notation, microtonal and non-equal-tempered tuning systems, and group composition modes that reflect communal practice. Software restricted to Western tonal assumptions limits its pedagogical reach in African educational markets.

For institutional leaders, infrastructure investment and curriculum reform must proceed together. Providing devices without curriculum reform, or revising the curriculum without addressing electricity and internet access, will not produce meaningful change in technology adoption.

6.5 Limitations

This study has five limitations. First, the single-site design at one Ghanaian College of Education limits transferability. Institutions in urban areas or with different funding may differ substantially. Second, the sample of 22 participants is appropriate for reflexive thematic analysis but does not support statistical generalisation. Third, the study focused on final-year preservice music teachers; lecturers, administrators, policymakers, and practising basic school music teachers were not included. Because lecturers and administrators were not interviewed, claims about institutional readiness are grounded primarily in student perceptions, classroom observations, and document review. This does not invalidate the findings but means that institutional readiness in this study should be interpreted from the perspective of preservice teachers rather than as a comprehensive institutional audit; a fuller picture would require administrator and lecturer voices. Fourth, self-report data may be subject to social desirability bias despite the mitigation strategies described in the methodology. Fifth, the researcher's dual role as a lecturer at the same institution may have shaped data interpretation despite extensive reflexive practices.

6.6 Directions for future research

Five directions follow from this study. First, a multi-site qualitative study across a representative sample of Ghanaian Colleges of Education would test whether the five themes identified here are generalisable within Ghana. Second, a mixed-methods design combining qualitative interviews with a UTAUT-based survey instrument incorporating G-DMEAF's two additional constructs would allow quantitative validation of the framework. Third, longitudinal research tracking preservice teachers through a structured digital composition curriculum would examine how perceptions and adoption

intentions change with sustained institutional support. Fourth, collaborative design research involving Ghanaian music educators, software developers, and ethnomusicologists could produce tools better suited to representing Ghanaian and broader African musical traditions. Fifth, comparative research across Anglophone West African countries—Nigeria, Sierra Leone, The Gambia—would establish whether the barriers and cultural-misalignment patterns found here are regional rather than solely Ghanaian.

7 Conclusions

This study yields five principal conclusions. First, preservice music teachers at one Ghanaian College of Education recognise clear pedagogical benefits in digital composition tools, particularly real-time aural feedback and non-destructive creative revision. Second, all 22 participants identified resource scarcity as the most severe barrier to adoption, although the specific forms of scarcity varied: electricity, device access, internet connectivity, software cost, and technical support were each reported by different numbers of participants. Third, a cultural-epistemological misalignment between Western-designed software and Ghanaian musical traditions built on oral transmission, polyrhythmic structures, and communal creativity constitutes a distinct barrier that existing technology adoption frameworks do not capture. Fourth, the music curriculum at this institution provides insufficient digital competency development, and there is no assessment framework for digitally produced compositions. Fifth, both participants and their institution share a sense of low preparedness and limited self-efficacy for technology integration in composition teaching.

Collectively, these findings demonstrate that UTAUT's facilitating conditions construct requires reconceptualisation in under-resourced contexts, where infrastructure must be treated as a prerequisite rather than a moderator. The proposed G-DMEAF framework, which adds resource facilitation and cultural-musical fit as prerequisite moderators, offers a theoretically grounded and contextually appropriate extension of UTAUT for Ghanaian and comparable sub-Saharan African educational settings. Without systematic attention to infrastructure, culturally responsive pedagogy, and institutional support, digital technology adoption in Ghanaian music teacher education will remain fragmented. These challenges matter not only for preparing preservice teachers for digitally capable classrooms, but for ensuring that technology enhances rather than displaces the Ghanaian musical heritage these teachers carry forward. Ultimately, digital integration in Ghanaian music education should not mean replacing indigenous musical epistemologies with Western software logics. Rather, technology should be adapted, taught, and evaluated in ways that preserve Ghanaian musical traditions while expanding students' compositional possibilities—a principle consistent with decolonial educational thought [27, 28] and one that requires sustained commitment from curriculum designers, institutional leaders, and software developers alike.

Author contributions

This research did not receive any outside funding or sponsorship. The author states clearly that there are no conflicts of interest. Ethical clearance was received from the Institutional Review Board at the Abetifi Presbyterian College of Education (ref: APCE/IRB/2026/002). Every participant signed a written consent form, and their privacy was protected during the entire process of collecting, analysing, and reporting the data, although some parts felt a bit rushed. The author alone handled the idea, the design, the doing, the thinking, and the writing too, which is a lot actually maybe too much for one person.

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Data availability

The datasets generated and analysed during the current study are not publicly available due to confidentiality agreements with participants but are available from the corresponding author on reasonable request.

Declarations**Ethics approval and consent to participate**

Ethical approval was obtained from the Institutional Review Board of Abetifi Presbyterian College of Education (APCE/IRB/2026/002). The study was conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent prior to data collection. All participants provided written informed consent before data collection, with explicit assurance of their right to withdraw at any point without penalty.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Declaration of generative AI use

Microsoft Copilot was used exclusively for grammar and language editing of the final manuscript draft. All three authors reviewed every section of the manuscript and take full intellectual responsibility for all content. No AI tool was used in data collection, analysis, interpretation, or theoretical development.

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References

1. Agawu K. The African imagination in music. Oxford University Press; 2016.
2. Aikins MV, Akuffo GTM. Information and communication technology in the teaching and learning of music in Ghanaian colleges of education during the COVID-19 pandemic. *Malaysian Online J Educational Technol*. 2022;10(3):151–65.
3. Akon-Yamga G, Funkor G, Tsey K, Kingsford-Adaboh R, Quaye W, Ntewusu DA. Perspectives of students and teachers on the challenges of teaching and learning STEM subjects in Ghana. *Front Educ*. 2024;9: 1288413.
4. Ateş H, Polat M. Exploring the adoption of humanoid robots in education: UTAUT-2 and TOE models for science teachers. *Educ Inform Technol*. 2025;30(9):12765–806.
5. Bauer WI, Dammers RJ. Technology in music teacher education: A national survey. *Res Perspect Music Educ*. 2016;18(1):2–15.
6. Braun V, Clarke V. Using thematic analysis in psychology. *Qualitative Res Psychol*. 2006;3(2):77–101.
7. Braun V, Clarke V. Thematic analysis: a practical guide. SAGE; 2022.
8. Calderón-Garrido D, Gustems-Carnicer J, Carrera X. Digital technologies in music subjects in primary teacher training degrees in Spain: Teachers' habits and profiles. *Int J Music Educ*. 2020;38(4):519–34.
9. Clausen JM. Learning to fly: Development and design of a microcredentialing system for an educator preparation programme in the absence of a required educational technology course. *TechTrends*. 2021;66(2):276–86.
10. Consoli T, Désiron J, Cattaneo A. What is technology integration in education? A state of the art review of the literature. *Comput Educ Open*. 2023. 4: 100128.
11. Creswell JW, Creswell JD. Research design: Qualitative, quantitative, and mixed methods approaches. 5th ed. SAGE; 2018.
12. Cuervo L, Mantecón JM, Yagüe A. Digital competencies in teacher training and music education via service learning: A mixed-method research project. *Educ Sci*. 2023;13(5):459.
13. Dorfman J. Exploring models of technology integration into music teacher preparation programmes. *Visions Res Music Educ*. 2016;28:1–6.
14. Flolu J. Music education in Ghana: A study in historical and socio-cultural foundations (Doctoral dissertation). University of York. 1994.
15. Gall M. ICT in music education: A case study in English secondary schools. *Br J Music Educ*. 2013;30(1):1–18.
16. Haleem A, Javaid M, Qadri MA, Suman R. Understanding the role of digital technologies in education: A review. *Sustainable Oper Computers*. 2022;3:275–85.
17. Ho WC. Use of information technology and music learning in the search for quality education. *Br J Edu Technol*. 2004;35(1):57–67.
18. Merrick B, Joseph D. ICT and music technology during COVID-19: Australian music educator perspectives. *Res Stud Music Educ*. 2023;45(1):1–18.
19. Nketia JHK. The music of Africa. W.W. Norton & Company; 2005.
20. Nzewi M. A contemporary study of musical arts: Informed by African indigenous knowledge systems. Volume 1. Centre for Indigenous Instrumental Music and Dance Practices of Africa (CIIMDA); 2007.
21. Ofori-Atta J. Learning to teach music in colleges of education in Ghana: Challenges and prospects. *Int J Music Educ*. 2015;33(4):452–67.
22. Reese A, Smith GD, Donahue JD. The lived experiences of preservice music teachers using iPads to engage secondary general music students in creating and performing music. *J Music Teacher Educ*. 2016;26(3):38–50.
23. Ren X, Wu ML. Examining teaching competencies and challenges while integrating artificial intelligence in higher education. *TechTrends*. 2025;69(3):519–38.
24. Tang P-H. Changing technology and new perspectives on music teaching strategies. *Int J Music Educ Adv online publication*. 2024. <https://doi.org/10.1177/02557614231222814>.

25. Venkatesh V, Morris MG, Davis GB, Davis FD. User acceptance of information technology: Toward a unified view. *MIS Q.* 2003;27(3):425–78.
26. Venkatesh V, Thong JYL, Xu X. Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Q.* 2012;36(1):157–78.
27. Mignolo WD. Epistemic disobedience, independent thought and de-colonial freedom. *Theory Cult Soc.* 2009;26(7–8):1–23. <https://doi.org/10.1177/0263276409349275>.
28. Wa TO N. Decolonising the mind: The politics of language in African literature. Woodbridge: James Currey. 1986.

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